

ADA / WCAG 2.1 Accessibility Audit

Page 2 — Colorado Springs 2C ArcGIS StoryMap

URL: storymaps.arcgis.com/stories/a05f42cbf32843a8b1458d3fd783e15a

Platform: Third-party (Esri ArcGIS StoryMaps)

Standard: WCAG 2.1 Level AA • **Legal frame:** ADA Title II

Audit Date: May 13, 2026

Auditor: Emma Moore, independent self-assessment

Summary

Twelve distinct accessibility findings were identified on the ArcGIS StoryMap page, with three rated **Critical**. The StoryMap is delivered through a third-party Esri platform, which means some markup is outside the City's direct control. However, under **ADA Title II**, the City retains accessibility responsibility for embedded third-party content used to deliver public information. Findings flagged below as platform-default issues should be raised with Esri support; the remaining findings can be addressed within the StoryMap configuration itself.

Severity	Count
Critical	3
High	4
Medium	3
Low / Advisory	2
Total	12

Findings

CRITICAL

1. Skip Link Target Does Not Exist

Element: Skip-to-content link pointing to #ref-n-onnRab

WCAG: 2.4.1 Bypass Blocks (Level A)

Impact: The page has a "Skip to content" link pointing to #ref-n-onnRab, but no element with that ID exists in the DOM. Keyboard-only users activating the skip link will not skip past the navigation — they remain at the top of the page.

Fix: Ensure id="ref-n-onnRab" is present on the main content container, or update the skip link's href to match an existing anchor.

2. 23 Images Have Empty Alt Text Without Being Marked Decorative

Element: 23 of 53 images on the StoryMap page

WCAG: 1.1.1 Non-text Content (Level A)

Impact: Twenty-three images have alt="" but are not marked role="presentation" or aria-hidden="true". Several appear to be content images (blob URLs and inline SVGs used as slide backgrounds and section dividers) rather than truly decorative.

Fix: For decorative images, add role="presentation" alongside alt="". For content images (such as the before/after slider frames), add descriptive alt text that communicates the content to screen reader users.

3. Malformed Link with Invalid URL

Element: Duryea Drive section: href="https://Duryea Drive repaved with 2C funds"

WCAG: 2.4.4 Link Purpose (Level A)

Impact: A link in the Duryea Drive section has a completely invalid URL — a description was placed in the href field. The link text reads "Open link" with no aria-label, and the destination is non-functional.

Fix: Either remove this link or correct the href to point to the actual resource (likely a photo or project page). Add descriptive text beyond "Open link."

HIGH

4. Broken Heading Hierarchy — H1 Followed by H4

Element: First headings after the page H1

WCAG: 1.3.1 Info and Relationships (Level A)

Impact: After the H1 ("Colorado Springs 2C Road Improvements"), the next headings are H4s, followed by H3s, then H2s. This skips directly from H1 to H4, a significant structural violation that disrupts document outline navigation.

Fix: Remap heading levels so H1 is followed by H2 section titles, H3 subsections, and H4 for any further sub-divisions. The first two H4 text-block headings should likely be removed or converted to <p> elements, as they read like paragraph leads rather than section headers.

5. Empty Heading Element

Element: One <h3> with empty text content

WCAG: 2.4.6 Headings and Labels (Level AA)

Impact: An H3 element exists in the DOM with completely empty text content. This creates a phantom heading in the document outline that screen readers may announce as "heading level 3" with no label.

Fix: Remove or replace the empty heading. If it serves as a visual spacer, use CSS instead.

6. Video Caption Tracks Have No Source URL

Element: Two <track kind="captions"> elements with src=null

WCAG: 1.2.2 Captions, Prerecorded (Level A)

Impact: Both video elements have a `<track kind="captions">` present (good), but the `src` attribute is null — no caption file is actually loaded. Captions appear supported in the DOM but are not available to users.

Fix: Provide a valid .vtt caption file URL in the `src` attribute of each `<track kind="captions">` element.

7. "More Actions" Menu Button Has No Accessible Label

Element: Top-navigation "■" button

WCAG: 4.1.2 Name, Role, Value (Level A)

Impact: The "■" menu button has `aria-expanded="false"` (correct) but no `aria-label`. Screen readers announce it only as "button" with no indication of its purpose.

Fix: Add `aria-label="More actions"` to this button element.

MEDIUM

8. Video Player Application Regions Lack ARIA Labels

Element: Two containers with `role="application"`

WCAG: 4.1.2 Name, Role, Value (Level A)

Impact: Two video player widget containers use `role="application"` without an `aria-label`. Assistive technologies switch to "application" interaction mode when encountering these regions, and users need to know what application they are entering.

Fix: Add `aria-label="2C Road Improvements overview video player"` (or similar specific label) to both application containers.

9. Map Controls Use Generic Elements Instead of Buttons (Esri platform issue)

Element: Zoom In, Zoom Out, Reset Orientation, Default View

WCAG: 4.1.2 Name, Role, Value (Level A)

Impact: The embedded map's zoom controls are rendered as `<generic>` elements with `type="button"` rather than actual `<button>` elements. This can cause inconsistent behavior with keyboard navigation and assistive technologies. This is an Esri/ArcGIS platform issue that requires configuration change or a support request to Esri.

Fix: File a support request with Esri to surface this as a platform accessibility issue. In the interim, document the limitation in the City's accessibility statement.

10. "Open Live Content" Link Lacks Context

Element: Link near embedded map

WCAG: 2.4.4 Link Purpose (Level A)

Impact: A link near the embedded map reads only "Open live content in a new tab" with no additional context about what the content is. Users navigating by links have no way to understand the destination without reading surrounding content.

Fix: Revise to "Open 2C Paving Map in a new tab" or add an `aria-label` providing context.

LOW / ADVISORY

11. Mobile Orientation Instruction Not Programmatically Associated

Element: "When using a mobile device, rotate to Landscape..."

WCAG: 1.3.3 Sensory Characteristics (Level A)

Impact: The orientation instruction is only visible as plain text and is not associated with the map element it refers to. Instructions using sensory characteristics (device orientation) should be associated with the relevant content.

Fix: Use `aria-describedby` on the map element pointing to the instruction's ID, or place the instruction immediately before the map in DOM order.

12. Before/After Image Sliders Lack Visible Instructions

Element: Six before/after image comparison sliders

WCAG: 2.5.3 Label in Name (Level A) — advisory

Impact: While the slider thumb has an accessible label ("Compare media by using the left and right arrow keys"), this instruction is only available to those who discover the slider via keyboard. Visual and touch users may not understand the interaction model.

Fix: Add a brief visible instruction such as "Drag or use arrow keys to compare before and after" near each slider.

Quick-Reference Summary Table

#	Issue	WCAG	Severity
1	Skip link target ID missing from DOM	2.4.1	Critical
2	23 images with empty alt not marked decorative	1.1.1	Critical
3	Malformed/invalid link URL on Duryea Drive	2.4.4	Critical
4	H1 followed directly by H4 (skips H2, H3)	1.3.1	High
5	Empty H3 heading element	2.4.6	High
6	Caption track elements have null src	1.2.2	High
7	"More Actions" button no accessible label	4.1.2	High
8	Video player application regions lack labels	4.1.2	Medium
9	Map controls use generic elements (Esri)	4.1.2	Medium
10	"Open live content" link lacks context	2.4.4	Medium
11	Mobile orientation note not associated with map	1.3.3	Low
12	Before/after sliders lack visible instructions	2.5.3	Low

Recommended Priority Order for Remediation

1. Fix the broken skip-link target ID (critical assistive technology blocker)
2. Resolve the invalid Duryea Drive link URL
3. Audit and remediate the 23 empty-alt images
4. Repair the heading hierarchy starting after the H1
5. Remove the empty H3 heading element
6. Provide real .vtt caption files for both videos
7. Label the "More Actions" button
8. Add aria-label to both video player application regions
9. Add context to the "Open live content" link
10. File an Esri support request for the map-control element-type issue
11. Add visible instructions near before/after sliders
12. Programmatically associate the mobile-orientation note with the map

Note on Third-Party Platform Responsibility

Because the StoryMap is delivered through Esri ArcGIS, several findings (most notably the map-control element-type issue) may require configuration changes within the Esri platform or support requests to Esri rather than direct code changes by the City. Under ADA Title II, however, the City remains responsible for the accessibility of public-facing content regardless of the platform on which it is delivered. The City should evaluate whether the StoryMap meets its ADA obligations and, where platform constraints prevent compliance, document alternative-access provisions in the City's accessibility statement.

Notes on Methodology

This audit is a self-assessment conducted by the author against the WCAG 2.1 Level AA standard, which is the conformance target referenced in the City of Colorado Springs Web Accessibility statement. Findings were identified through inspection of the published page source and accessibility tree. Because this page is hosted on a third-party platform (Esri ArcGIS StoryMaps), some findings reflect platform-default markup rather than configurable content. Under ADA Title II, the City retains accessibility responsibility for embedded third-party content used to deliver public information. A formal third-party audit conducted by an IAAP-credentialed auditor (CPACC or WAS) or by an accessibility firm (Deque, Level Access, TPGi, Knowbility, or equivalent) is recommended prior to any binding conformance claim. This document is offered as a constructive starting point for the City's web team and Office of Accessibility.

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